

Azure Cloud Fundamentals and Data Pipeline Implementation

Using Azure Data Factory (ADF)

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Objective

The objective of this assignment is to understand Microsoft Azure cloud services and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory. The assignment includes creating Azure resources, storing data in Blob Storage, building a data pipeline, validating metadata, assigning IAM roles, and monitoring pipeline execution.

Tools and Services Used

Service	Purpose
Microsoft Azure Portal	Cloud resource management
Azure Resource Group	Logical grouping of resources
Azure Storage Account	Cloud storage service
Azure Blob Storage	Store CSV files
Azure Data Factory	Build and execute data pipelines
Azure IAM	Access management and permissions

Task 1 – Explore Azure Portal

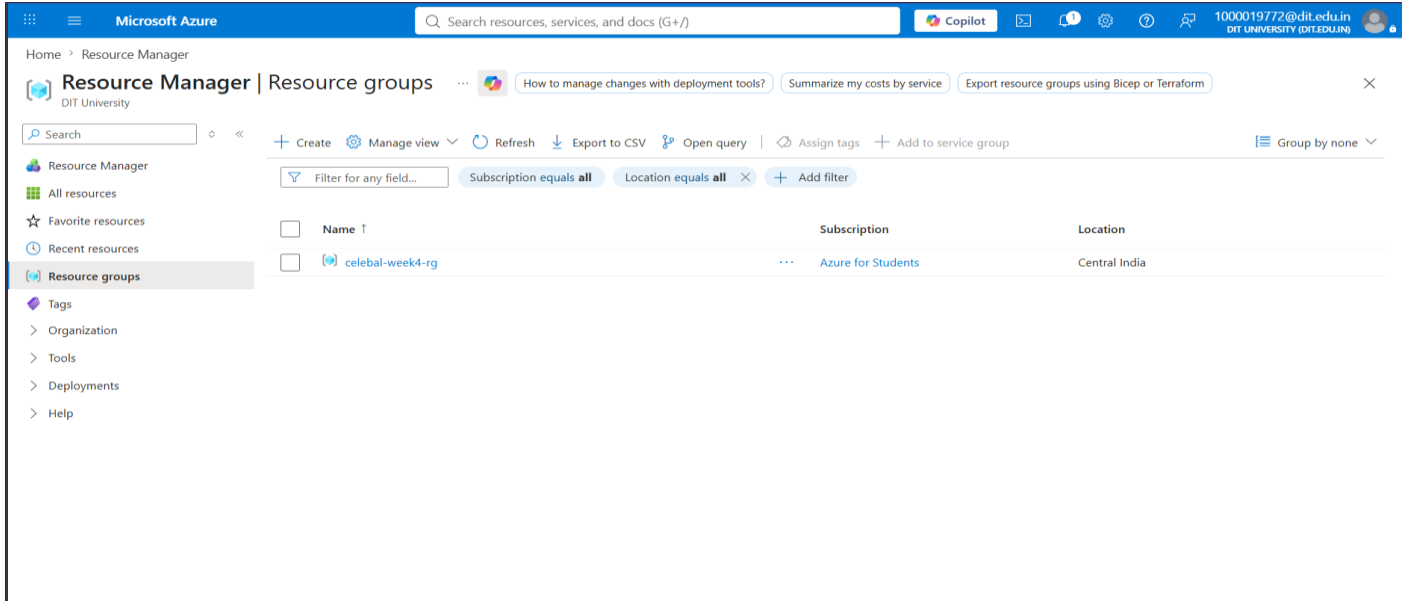
Objective

Create a Resource Group to organize all Azure resources used during the assignment.

Procedure

1. Logged into the Microsoft Azure Portal.
2. Opened the **Resource Groups** service.
3. Created a new Resource Group named **celebal-week4-rg**.
4. Selected **Central India** as the deployment region.
5. Verified successful creation of the Resource Group.

Screenshot:



Task 2 – Storage Setup

Objective

Create an Azure Storage Account, configure Blob Storage containers, and upload the source CSV dataset.

Procedure

Step 1: Create Storage Account

1. Opened Azure Storage Accounts.
2. Created a storage account named: **celebalweek4storageead11**
3. Selected:
 - Resource Group: **celebal-week4-rg**
 - Region: **Central India**
 - Performance: Standard
 - Replication: Locally Redundant Storage (LRS)

Screenshot:

The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows the user is logged in as '1000019772@dit.edu.in'. The main content area shows the 'Overview' page for the storage account 'celebalweek4storageead11'. The left sidebar contains a navigation menu with options like 'Overview', 'Activity log', 'Tags', 'Diagnose and solve problems', 'Access Control (IAM)', 'Data migration', 'Events', 'Storage browser', 'Storage Mover', 'Partner solutions', 'Resource visualizer', 'Data storage', 'Security + networking', 'Data management', 'Settings', 'Monitoring', and 'Automation'. The main content area is divided into sections: 'Essentials' (Resource group, Location, Subscription ID, Subscription ID, Disk state, Tags), 'Properties' (Blob service settings like Hierarchical namespace, Default access tier, Blob anonymous access, Blob soft delete, Container soft delete, Versioning, Change feed, NFS v3, Allow cross-tenant replication), 'Security' (Require secure transfer for REST API operations, Storage account key access, Minimum TLS version, Infrastructure encryption), and 'Networking' (Public network access, Public network access scope, Private endpoint connections).

Step 2: Create Blob Containers

Three Blob Containers were created: \$logs , input, output

Screenshot:

The screenshot displays the Microsoft Azure portal interface, showing the 'Containers' page for the storage account 'celebalweek4storageead11'. The left sidebar contains a navigation menu with options like 'Overview', 'Activity log', 'Tags', 'Diagnose and solve problems', 'Access Control (IAM)', 'Data migration', 'Events', 'Storage browser', 'Storage Mover', 'Partner solutions', 'Resource visualizer', 'Data storage', 'Security + networking', and 'Data management'. The main content area shows a table of containers. The table has columns: 'Name', 'Last modified', 'Anonymous access level', 'Lease state', and 'Storage connector'. Three containers are listed: '\$logs', 'input', and 'output'. All containers are in 'Available' state and have 'Private' anonymous access level.

Name	Last modified	Anonymous access level	Lease state	Storage connector
\$logs	12/07/2026, 15:04:55	Private	Available	-
input	12/07/2026, 15:16:38	Private	Available	-
output	12/07/2026, 15:17:44	Private	Available	-

Step 3: Upload Source Dataset

The CSV file:

Sample - Superstore.csv was uploaded into the **input** container.

The uploaded dataset serves as the source file for subsequent Azure Data Factory operations.

Screenshot:

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

1000019772@dit.edu.in
DIT UNIVERSITY (DIT.EDU.IN)

Home > celebalweek4storagead11_1783848544556 | Overview > celebalweek4storagead11 | Containers

input

Container

Search

+ Add Directory

↑ Upload

🔒 Change access level

🔄 Refresh

🗑 Delete

📄 Copy

📄 Paste

🔄 Rename

🔑 Acquire lease

🔑 Break lease

🔧 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

input

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	Sample - Superstore.csv	12/07/2026, 15:19:09	Hot (Inferred)	Block blob	2.18 MiB	Available	⋮

Task 3 – Azure Data Factory Basics

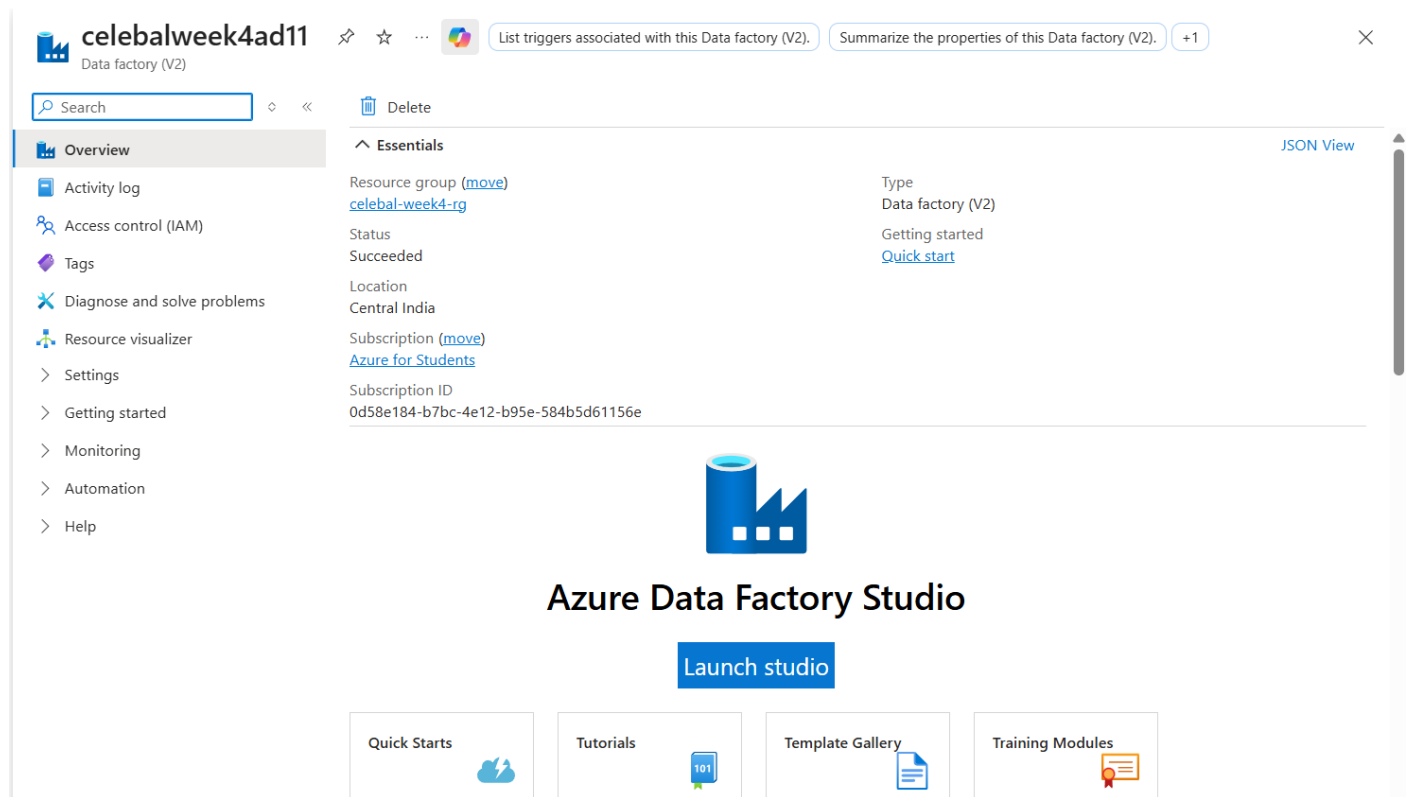
Objective

Create Azure Data Factory and configure all components required for building data pipelines.

Step 1: Create Azure Data Factory

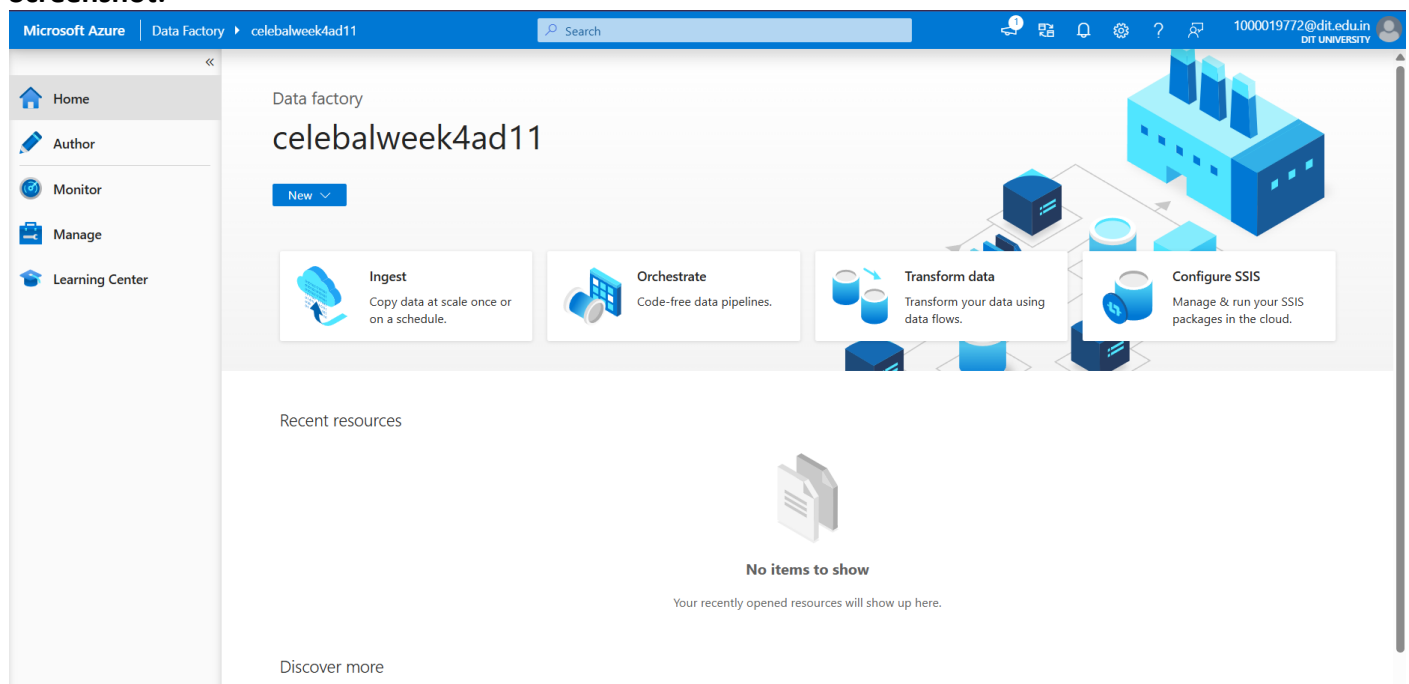
A new Azure Data Factory instance named **celebalweek4ad11** was created inside the existing Resource Group.

Screenshot:



After deployment, Azure Data Factory Studio was launched for authoring and monitoring pipelines.

Screenshot:

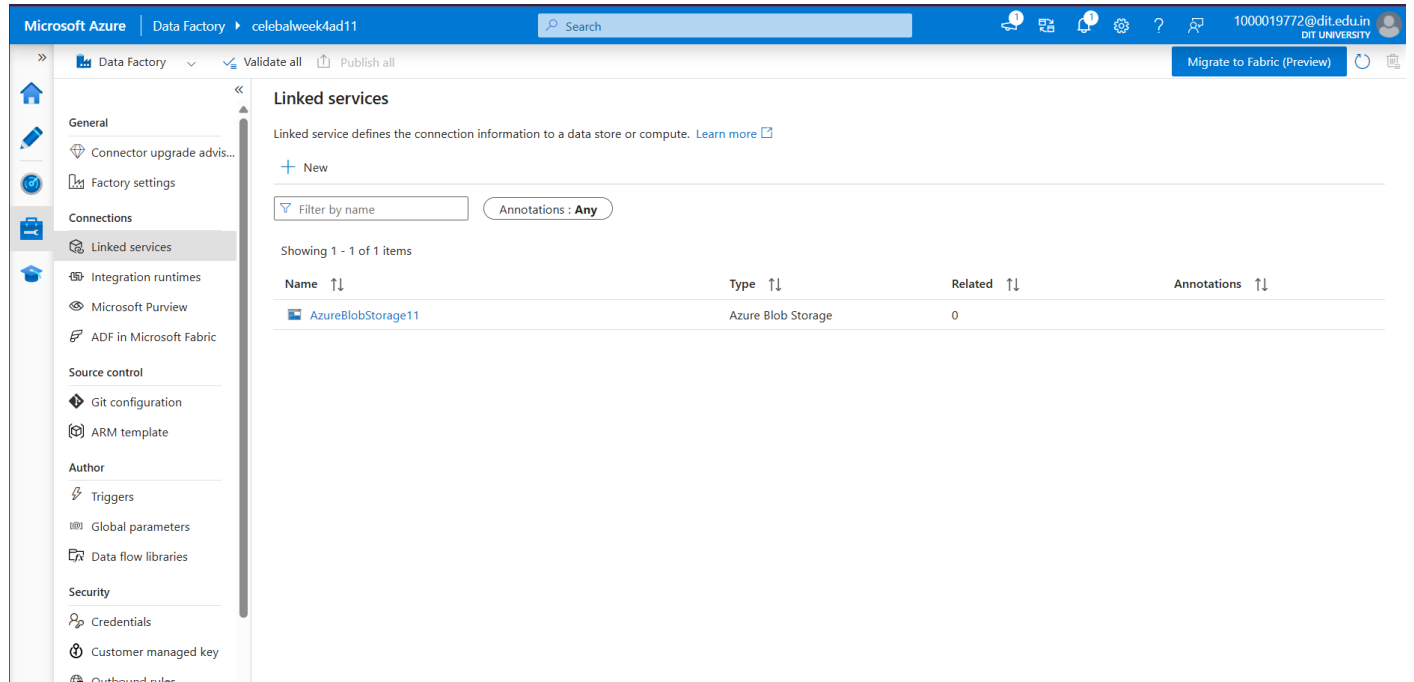


Step 2: Create Linked Service

A Linked Service establishes the connection between Azure Data Factory and Azure Blob Storage.

A linked service named **AzureBlobStorage11** was created using Azure Blob Storage authentication.

Screenshot:



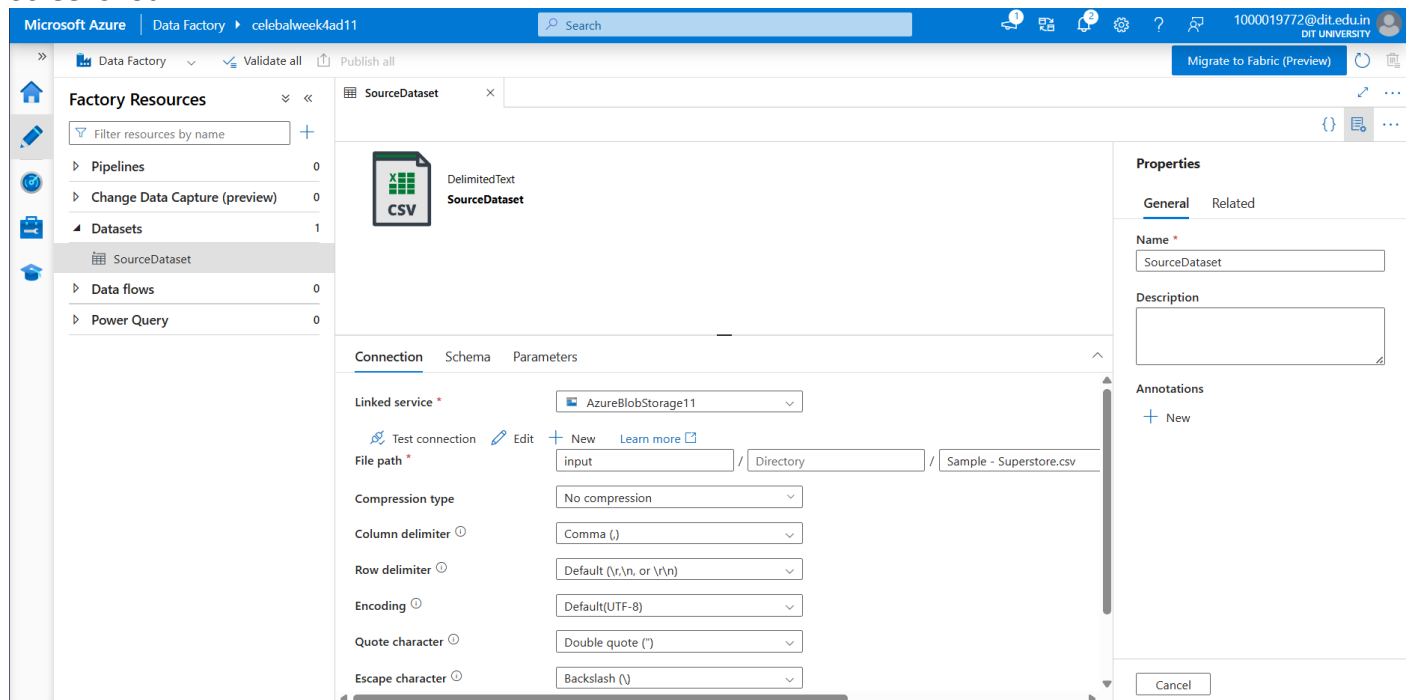
Step 3: Create Source Dataset

A dataset named **SourceDataset** was created with the following configuration:

- Linked Service: AzureBlobStorage11
- Container: input
- File: Sample - Superstore.csv
- File Type: Delimited Text (CSV)

This dataset represents the source data used by the pipelines.

Screenshot:



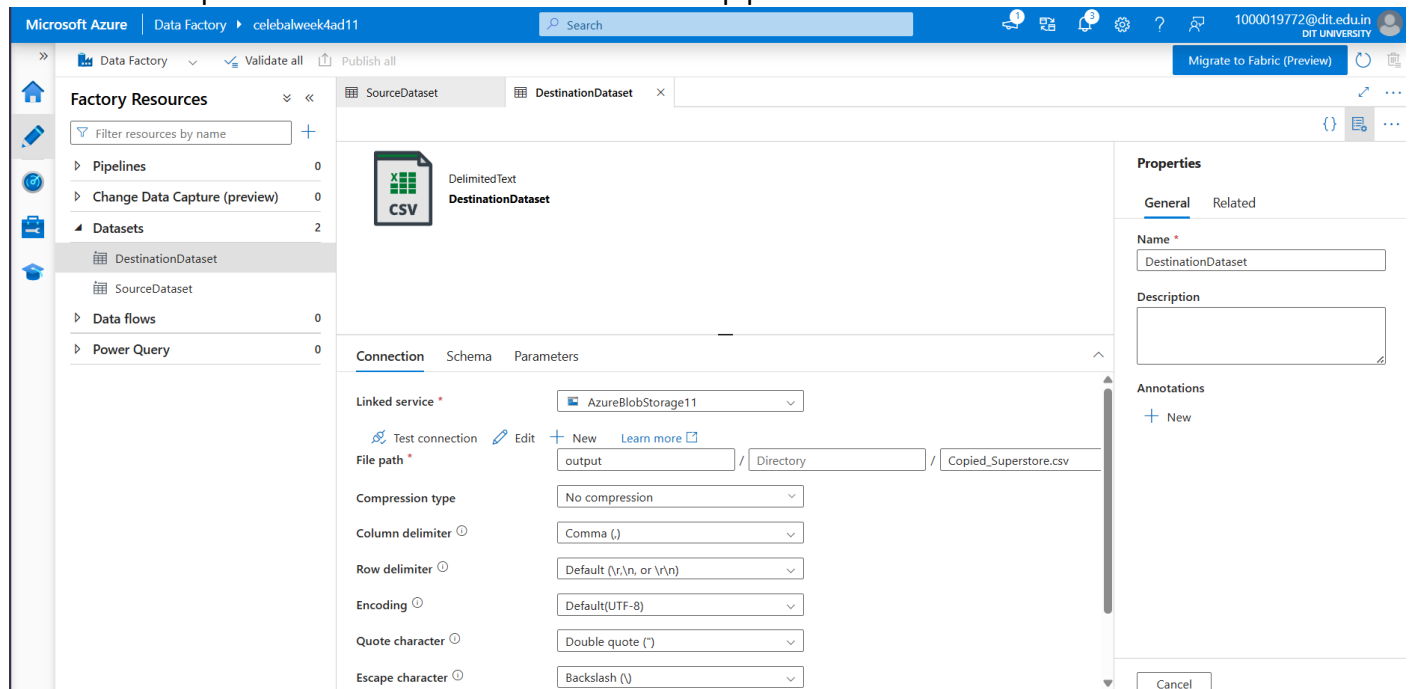
Step 4: Create Destination Dataset

A second dataset named **DestinationDataset** was created for storing copied data.

Configuration:

- Linked Service: AzureBlobStorage11
- Container: output
- File: Copied_Superstore.csv

This dataset represents the destination location of the pipeline.



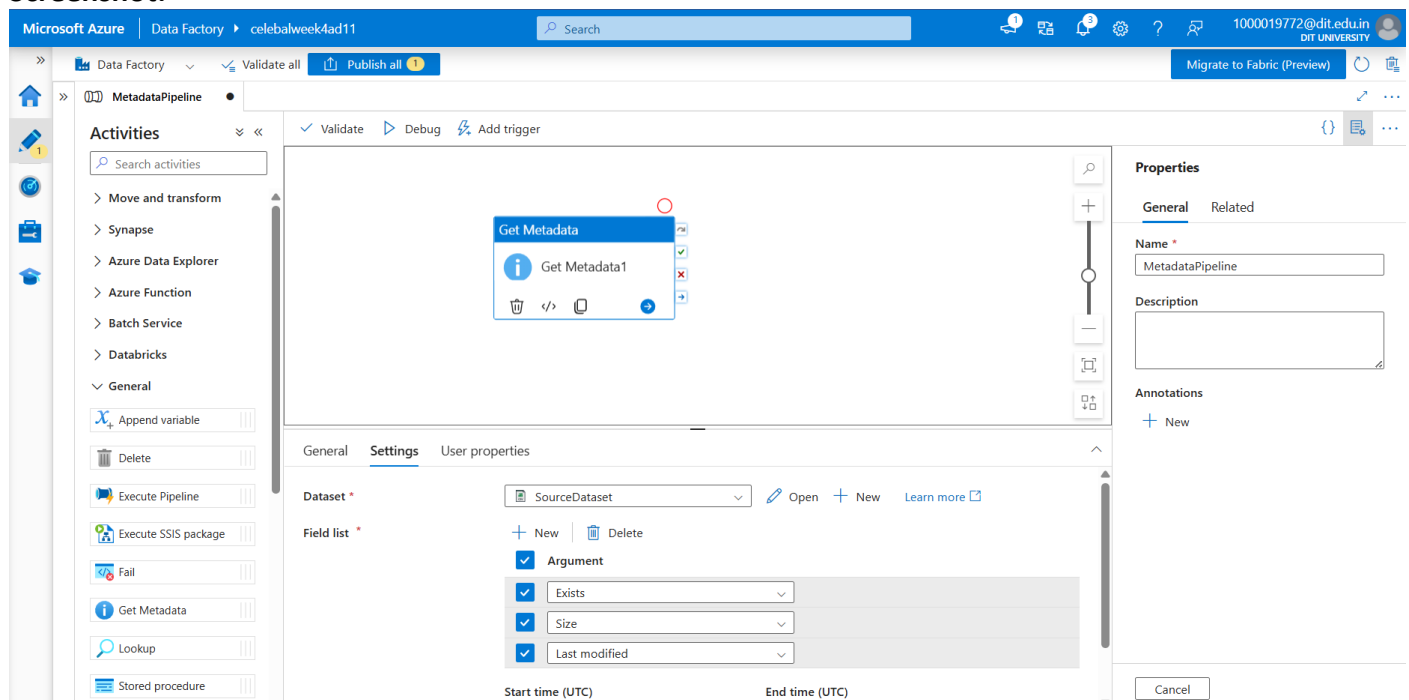
Step 5: Create Metadata Pipeline

A pipeline named **MetadataPipeline** was created using the **Get Metadata** activity.

The SourceDataset was selected, and the following metadata properties were configured:

- Exists
- Size
- Last Modified

Screenshot:



The Metadata Pipeline executed successfully and validated the existence and properties of the source dataset.

The Metadata Pipeline executed successfully and validated the existence and properties of the source dataset.

The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar includes the Microsoft Azure logo, the text 'Data Factory', and the user's email 'celebalweek4ad11'. A search bar is located on the right side of the top bar. The main interface is divided into three sections: a left sidebar, a central workspace, and a right sidebar.

Left Sidebar: Contains navigation icons for Home, Recent, and a list of data sources: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, and General. Below these are activity templates: Append variable, Delete, Execute Pipeline, Execute SSIS package, Fail, Get Metadata, Lookup, and Stored procedure.

Central Workspace: Displays the 'MetadataPipeline' pipeline. The top bar shows 'Validate', 'Debug', and 'Add trigger' buttons. Below this is a visual representation of the pipeline with a 'Get Metadata' activity. The 'Output' tab is selected, showing the pipeline run ID '2869f0bf-0481-4ba7-a0d6-7184f54a462a' and the status 'Succeeded'. A table lists the activity details:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Get Metadata1	Succeeded	Get Metadata	7/12/2026, 4:39:19 PM	15s	AutoResolveIntegrationRun

Right Sidebar: Contains the 'Properties' section with 'General' and 'Related' tabs. The 'Name' field is set to 'MetadataPipeline'. Below this is a 'Description' field and an 'Annotations' section with a '+ New' button. A 'Cancel' button is located at the bottom right.

Task 4 – Pipeline Development

Objective

Develop a pipeline to copy the source CSV file from one Blob Storage container to another.

Procedure

A pipeline named **CopyPipeLine** was created using the **Copy Data** activity.

Configuration:

Source:

- Dataset: SourceDataset

Destination:

- Dataset: DestinationDataset

The Copy Data activity transfers the CSV file from the **input** container to the **output** container.

The pipeline was validated successfully before execution.

Screenshot:

The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows the user is logged in as 1000019772@dit.edu.in. The main area shows the 'CopyPipeLine' pipeline configuration. The 'Activities' pane on the left lists 'Copy data' and 'Data flow'. The 'Copy data' activity is selected, and its configuration is shown in the center. The 'Output' tab is active, displaying the pipeline run ID 'eb57c250-a316-4e16-936c-db0fe9fef1ff' and the status 'Succeeded'. A table below shows the execution details for the 'CopySuperstoreFile' activity.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
CopySuperstoreFile	✓ Succeeded	Copy data	7/12/2026, 5:49:22 PM	14s	AutoResolveIntegrationRuntr

Task 5 – Pipeline Execution

Objective

Execute the developed pipeline and verify successful completion.

Procedure

1. Opened CopyPipeLine.
2. Clicked **Debug** to execute the pipeline.
3. Monitored execution status.
4. Confirmed successful completion.
5. Verified that the destination file was generated inside the output container.

Result

The pipeline completed successfully.

The file **Copied_Superstore.csv** was created in the destination Blob Storage container.

Screenshot:

The screenshot displays the Microsoft Azure portal interface. At the top, the navigation bar shows 'Microsoft Azure' and a search bar. Below the navigation bar, the breadcrumb trail indicates the location: 'Home > Storage center | Blob Storage > celebalweek4storagead11 | Containers'. The main content area shows the 'output' container. On the left, there is a sidebar with 'Overview' selected, and options for 'Diagnose and solve problems', 'Access Control (IAM)', and 'Settings'. The main area displays the 'output' container with a search bar and a list of actions: 'Add Directory', 'Upload', 'Change access level', 'Refresh', 'Delete', 'Copy', 'Paste', 'Rename', 'Acquire lease', 'Break lease', and 'Edit columns'. Below the actions, the authentication method is 'Access key (Switch to Microsoft Entra user account)'. A search bar for blobs by prefix is present. The table shows 'Showing all 1 items' with columns: 'Name', 'Last modified', 'Access tier', 'Blob type', 'Size', and 'Lease state'. The single item listed is 'Copied_Superstore.csv' with a last modified date of '12/07/2026, 17:49:33', an access tier of 'Hot (Inferred)', a blob type of 'Block blob', a size of '2.18 MiB', and a lease state of 'Available'.

Name	Last modified	Access tier	Blob type	Size	Lease state
Copied_Superstore.csv	12/07/2026, 17:49:33	Hot (Inferred)	Block blob	2.18 MiB	Available

Task 6 – IAM Roles

Objective

Configure Azure Identity and Access Management permissions.

Procedure

The following roles were configured:

- Owner
- Reader
- App Configuration Contributor

These permissions ensured appropriate access to the Storage Account and Azure Data Factory resources throughout the implementation.

Screenshot - Initial IAM Role Assignments:

The screenshot shows the 'Access Control (IAM)' page for the storage account 'celebalweek4storagead11'. The 'Role assignments' tab is selected, showing 2 role assignments for this subscription. The table lists the following assignments:

Name	Type	Role	Scope	Condition
Owner (1)				
Aditya Kumar (1000019772@dit.edu.in)	User	Owner	Subscription (Inherited)	None
Reader (1)				
Aditya Kumar (1000019772@dit.edu.in)	User	Reader	This resource	None

Showing 1 - 2 of 2 results.

Screenshot - Updated IAM Role Assignments:

The screenshot shows the 'Access Control (IAM)' page for the storage account 'celebalweek4storagead11'. The 'Role assignments' tab is selected, showing 3 role assignments for this subscription. The table lists the following assignments:

Name	Type	Role	Scope	Condition
Owner (1)				
Aditya Kumar (1000019772@dit.edu.in)	User	Owner	Subscription (Inherited)	None
Reader (1)				
Aditya Kumar (1000019772@dit.edu.in)	User	Reader	This resource	None
App Configuration Contributor (1)				
Aditya Kumar (1000019772@dit.edu.in)	User	App Configuration Contributor	This resource	None

Showing 1 - 3 of 3 results.

Mini Project

Problem Statement

Develop a complete Azure Data Factory solution capable of:

- Reading a CSV file from Azure Blob Storage
- Validating file metadata
- Copying the file to another Blob Storage location
- Successfully executing the workflow

Implementation

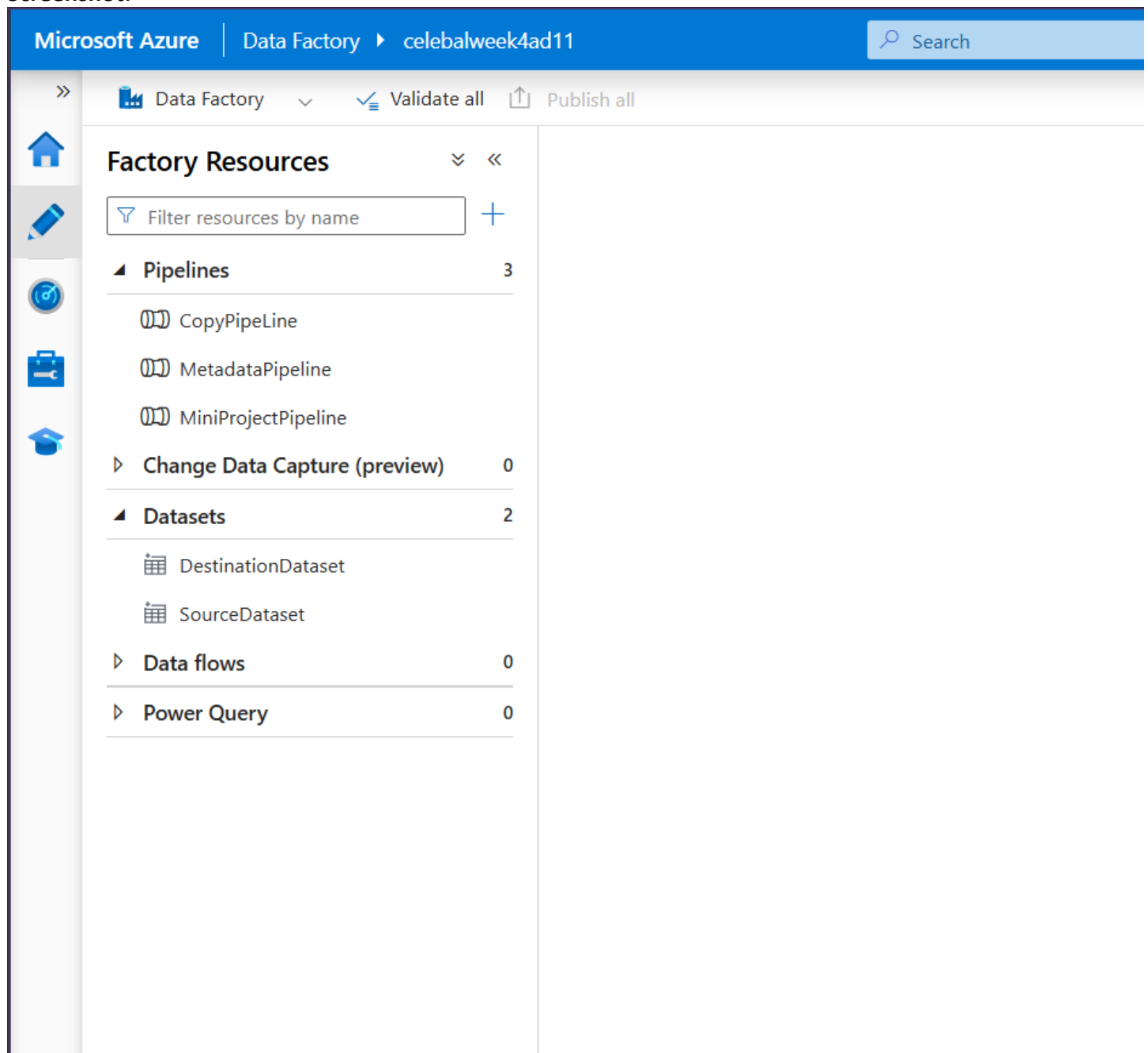
A pipeline named **MiniProjectPipeline** was developed to combine metadata validation and file copying into a single automated workflow.

Pipeline sequence:

1. Validate source file using **Get Metadata**
2. Upon successful validation, execute **Copy Data**
3. Store copied file in destination Blob Storage

The workflow follows a success dependency, ensuring that file copying occurs only after metadata validation completes successfully.

Screenshot:



Pipeline Execution

The MiniProjectPipeline was executed using Debug mode.

Execution Results:

- Metadata Validation – Succeeded
- Copy Data – Succeeded

The overall pipeline completed successfully without errors.

MiniProjectPipeline

Activities

Move and transform

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Validate

Debug

Add trigger

Get Metadata

Copy data

Parameters

Variables

Settings

Output

Pipeline run ID

ae9bca9c-8ac1-4948-9c28-de621fece399

Pipeline status

Succeeded

View debug run consumption

Monitor in Azure Metrics

Export to CSV

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run
CopyCSV	Succeeded	Copy data	7/12/2026, 7:26:46 PM	15s	AutoResolveIntegrationRuntime (Central India)		e43e384a-1
VaildateMetadata	Succeeded	Get Metadata	7/12/2026, 7:26:33 PM	12s	AutoResolveIntegrationRuntime (Central India)		af5b92e1-3

Output Verification

After successful execution, the output container contained **Copied_Superstore.csv** confirming that the pipeline correctly copied the source dataset.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

1000019772@dit.edu.in

DIT UNIVERSITY (DIT.EDU.IN)

Home

Storage center

Blob Storage

celebalweek4storageead11

Containers

output

Container

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

Authentication method: Access key

Switch to Microsoft Entra user account

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Copied_Superstore.csv	12/07/2026, 19:10:03	Hot (Inferred)	Block blob	2.18 MiB	Available

Conclusion

This assignment provided practical exposure to Microsoft's Azure cloud platform and Azure Data Factory. Beginning with the creation of cloud resources, Blob Storage was configured to store source and destination datasets, while Azure Data Factory was used to orchestrate metadata validation and file copying through reusable pipelines. Appropriate IAM permissions were configured to ensure secure access to cloud resources. The final mini project successfully integrated metadata validation and data copying into a single automated workflow, demonstrating the fundamental concepts of cloud storage, data integration, workflow orchestration, and pipeline execution. The successful completion of all tasks provided hands-on experience with Azure services commonly used in real-world cloud data engineering and ETL solutions.